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COMMON FORMULAS

Temperature

$$F = \frac{9}{5}C + 32$$

F = degrees Fahrenheit

C = degrees Celsius

Distance

$$d = rt$$

d = distance traveled

t = time

r = rate

Simple Interest

$$I = Prt$$

I = interest

P = principal

r = annual interest rate

t = time in years

Compound Interest

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

A = balance

P = principal

r = annual interest rate

n = compoundings per year

t = time in years

Coordinate Plane: Midpoint Formula

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

midpoint of line segment
joining (x_1, y_1) and (x_2, y_2)

Coordinate Plane: Distance Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

d = distance
between points
 (x_1, y_1) and (x_2, y_2)

Quadratic Formula

If $p(x) = ax^2 + bx + c$, $a \neq 0$ and $b^2 - 4ac \geq 0$, then
the real zeros of p are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

CONVERSIONS

Length and Area

1 foot = 12 inches

1 mile = 5280 feet

1 kilometer = 1000 meters

1 kilometer $\approx$ 0.621 mile

1 meter $\approx$ 3.281 feet

1 foot $\approx$ 0.305 meter

1 yard = 3 feet

1 mile = 1760 yards

1 meter = 100 centimeters

1 mile $\approx$ 1.609 kilometers

1 meter $\approx$ 39.370 inches

1 foot $\approx$ 30.480 centimeters

1 meter = 1000 millimeters

1 centimeter $\approx$ 0.394 inch

1 inch $\approx$ 2.540 centimeters

1 acre = 4840 square yards

1 square mile = 640 acres

Volume

1 gallon = 4 quarts

1 gallon = 231 cubic inches

1 liter = 1000 milliliters

1 liter $\approx$ 1.057 quarts

1 gallon $\approx$ 3.785 liters

1 quart = 2 pints

1 gallon $\approx$ 0.134 cubic foot

1 liter = 100 centiliters

1 liter $\approx$ 0.264 gallon

1 quart $\approx$ 0.946 liter

1 pint = 16 fluid ounces

1 cubic foot $\approx$ 7.48 gallons

Weight and Mass on Earth

1 ton = 2000 pounds

1 kilogram $\approx$ 2.205 pounds

1 pound = 16 ounces

1 pound $\approx$ 0.454 kilogram

1 kilogram = 1000 grams

1 gram $\approx$ 0.035 ounce

FORMULAS FROM GEOMETRY

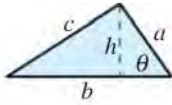
Triangle

$$h = a \sin \theta$$

$$\text{Area} = \frac{1}{2}bh$$

Laws of Cosines:

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

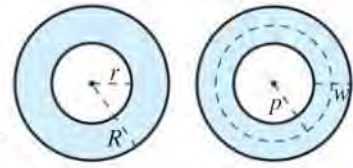


Circular Ring

$$\text{Area} = \pi(R^2 - r^2)$$

$$= 2\pi pw$$

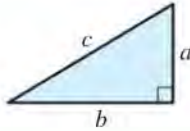
(p = average radius,
 w = width of ring)



Right Triangle

Pythagorean Theorem:

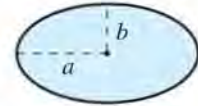
$$c^2 = a^2 + b^2$$



Ellipse

$$\text{Area} = \pi ab$$

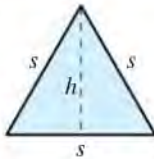
$$\text{Circumference} \approx 2\pi \sqrt{\frac{a^2 + b^2}{2}}$$



Equilateral Triangle

$$h = \frac{\sqrt{3}s}{2}$$

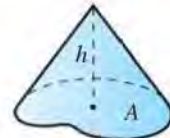
$$\text{Area} = \frac{\sqrt{3}s^2}{4}$$



Cone

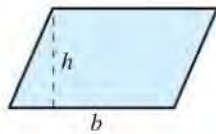
(A = area of base)

$$\text{Volume} = \frac{Ah}{3}$$



Parallelogram

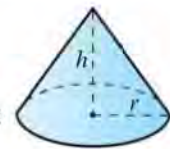
$$\text{Area} = bh$$



Right Circular Cone

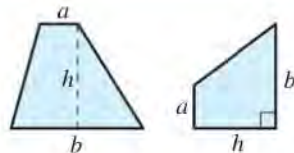
$$\text{Volume} = \frac{\pi r^2 h}{3}$$

$$\text{Lateral Surface Area} = \pi r \sqrt{r^2 + h^2}$$



Trapezoid

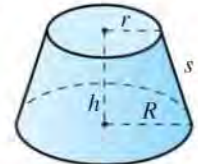
$$\text{Area} = \frac{h}{2}(a + b)$$



Frustum of Right Circular Cone

$$\text{Volume} = \frac{\pi(r^2 + rR + R^2)h}{3}$$

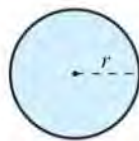
$$\text{Lateral Surface Area} = \pi s(R + r)$$



Circle

$$\text{Area} = \pi r^2$$

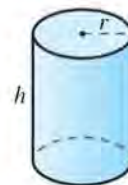
$$\text{Circumference} = 2\pi r$$



Right Circular Cylinder

$$\text{Volume} = \pi r^2 h$$

$$\text{Lateral Surface Area} = 2\pi rh$$

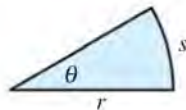


Sector of Circle

$$\text{Area} = \frac{\theta r^2}{2}$$

$$s = r\theta$$

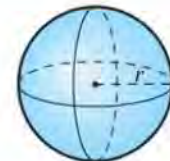
(θ in radians)



Sphere

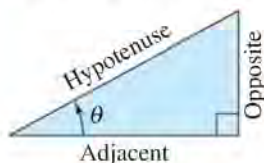
$$\text{Volume} = \frac{4}{3}\pi r^3$$

$$\text{Surface Area} = 4\pi r^2$$



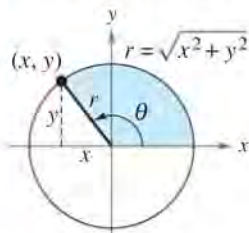
Definition of the Six Trigonometric Functions

Right triangle definitions, where $0 < \theta < \pi/2$



$$\begin{aligned}\sin \theta &= \frac{\text{opp.}}{\text{hyp.}} & \csc \theta &= \frac{\text{hyp.}}{\text{opp.}} \\ \cos \theta &= \frac{\text{adj.}}{\text{hyp.}} & \sec \theta &= \frac{\text{hyp.}}{\text{adj.}} \\ \tan \theta &= \frac{\text{opp.}}{\text{adj.}} & \cot \theta &= \frac{\text{adj.}}{\text{opp.}}\end{aligned}$$

Circular function definitions, where θ is any angle



$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$

Reciprocal Identities

$$\begin{aligned}\sin u &= \frac{1}{\csc u} & \cos u &= \frac{1}{\sec u} & \tan u &= \frac{1}{\cot u} \\ \csc u &= \frac{1}{\sin u} & \sec u &= \frac{1}{\cos u} & \cot u &= \frac{1}{\tan u}\end{aligned}$$

Quotient Identities

$$\tan u = \frac{\sin u}{\cos u} \quad \cot u = \frac{\cos u}{\sin u}$$

Pythagorean Identities

$$\begin{aligned}\sin^2 u + \cos^2 u &= 1 \\ 1 + \tan^2 u &= \sec^2 u & 1 + \cot^2 u &= \csc^2 u\end{aligned}$$

Cofunction Identities

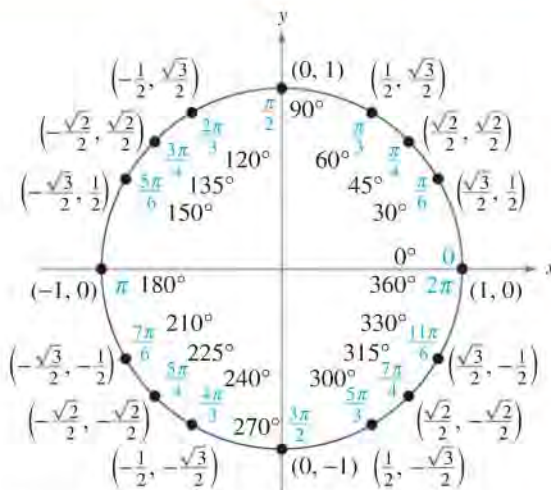
$$\begin{aligned}\sin\left(\frac{\pi}{2} - u\right) &= \cos u & \cot\left(\frac{\pi}{2} - u\right) &= \tan u \\ \cos\left(\frac{\pi}{2} - u\right) &= \sin u & \sec\left(\frac{\pi}{2} - u\right) &= \csc u \\ \tan\left(\frac{\pi}{2} - u\right) &= \cot u & \csc\left(\frac{\pi}{2} - u\right) &= \sec u\end{aligned}$$

Even/Odd Identities

$$\begin{aligned}\sin(-u) &= -\sin u & \cot(-u) &= -\cot u \\ \cos(-u) &= \cos u & \sec(-u) &= \sec u \\ \tan(-u) &= -\tan u & \csc(-u) &= -\csc u\end{aligned}$$

Sum and Difference Formulas

$$\begin{aligned}\sin(u \pm v) &= \sin u \cos v \pm \cos u \sin v \\ \cos(u \pm v) &= \cos u \cos v \mp \sin u \sin v \\ \tan(u \pm v) &= \frac{\tan u \pm \tan v}{1 \mp \tan u \tan v}\end{aligned}$$



Double-Angle Formulas

$$\begin{aligned}\sin 2u &= 2 \sin u \cos u \\ \cos 2u &= \cos^2 u - \sin^2 u = 2 \cos^2 u - 1 = 1 - 2 \sin^2 u \\ \tan 2u &= \frac{2 \tan u}{1 - \tan^2 u}\end{aligned}$$

Power-Reducing Formulas

$$\begin{aligned}\sin^2 u &= \frac{1 - \cos 2u}{2} \\ \cos^2 u &= \frac{1 + \cos 2u}{2} \\ \tan^2 u &= \frac{1 - \cos 2u}{1 + \cos 2u}\end{aligned}$$

Sum-to-Product Formulas

$$\begin{aligned}\sin u + \sin v &= 2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right) \\ \sin u - \sin v &= 2 \cos\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right) \\ \cos u + \cos v &= 2 \cos\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right) \\ \cos u - \cos v &= -2 \sin\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)\end{aligned}$$

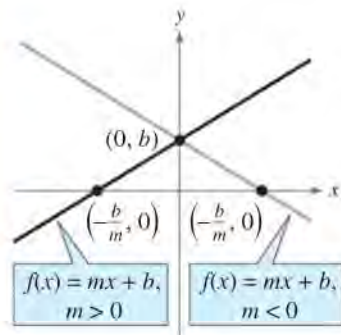
Product-to-Sum Formulas

$$\begin{aligned}\sin u \sin v &= \frac{1}{2} [\cos(u-v) - \cos(u+v)] \\ \cos u \cos v &= \frac{1}{2} [\cos(u-v) + \cos(u+v)] \\ \sin u \cos v &= \frac{1}{2} [\sin(u+v) + \sin(u-v)] \\ \cos u \sin v &= \frac{1}{2} [\sin(u+v) - \sin(u-v)]\end{aligned}$$

LIBRARY OF PARENT FUNCTIONS SUMMARY

Linear Function

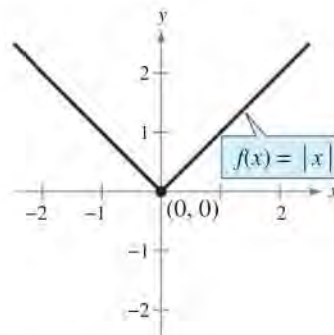
$$f(x) = mx + b$$



Domain: $(-\infty, \infty)$
 Range: $(-\infty, \infty)$
 x-intercept: $(-b/m, 0)$
 y-intercept: $(0, b)$
 Increasing when $m > 0$
 Decreasing when $m < 0$

Absolute Value Function

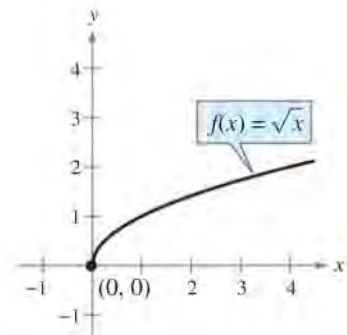
$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



Domain: $(-\infty, \infty)$
 Range: $[0, \infty)$
 Intercept: $(0, 0)$
 Decreasing on $(-\infty, 0)$
 Increasing on $(0, \infty)$
 Even function
 y-axis symmetry

Square Root Function

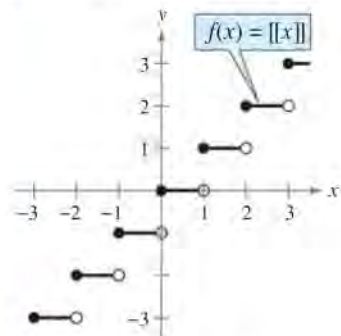
$$f(x) = \sqrt{x}$$



Domain: $[0, \infty)$
 Range: $[0, \infty)$
 Intercept: $(0, 0)$
 Increasing on $(0, \infty)$

Greatest Integer Function

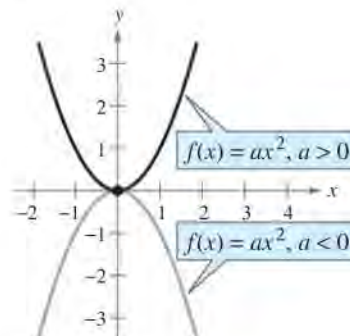
$$f(x) = \llbracket x \rrbracket$$



Domain: $(-\infty, \infty)$
 Range: the set of integers
 x-intercepts: in the interval $[0, 1)$
 y-intercept: $(0, 0)$
 Constant between each pair of consecutive integers
 Jumps vertically one unit at each integer value

Quadratic (Squaring) Function

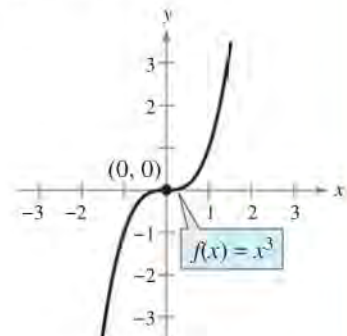
$$f(x) = ax^2$$



Domain: $(-\infty, \infty)$
 Range ($a > 0$): $[0, \infty)$
 Range ($a < 0$): $(-\infty, 0]$
 Intercept: $(0, 0)$
 Decreasing on $(-\infty, 0)$ for $a > 0$
 Increasing on $(0, \infty)$ for $a > 0$
 Increasing on $(-\infty, 0)$ for $a < 0$
 Decreasing on $(0, \infty)$ for $a < 0$
 Even function
 y-axis symmetry
 Relative minimum ($a > 0$),
 relative maximum ($a < 0$),
 or vertex: $(0, 0)$

Cubic Function

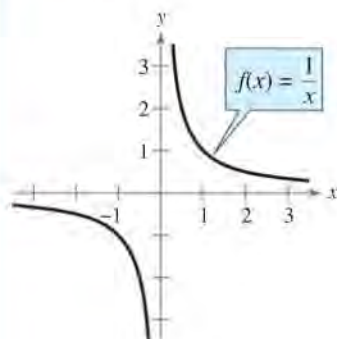
$$f(x) = x^3$$



Domain: $(-\infty, \infty)$
 Range: $(-\infty, \infty)$
 Intercept: $(0, 0)$
 Increasing on $(-\infty, \infty)$
 Odd function
 Origin symmetry

Rational (Reciprocal) Function

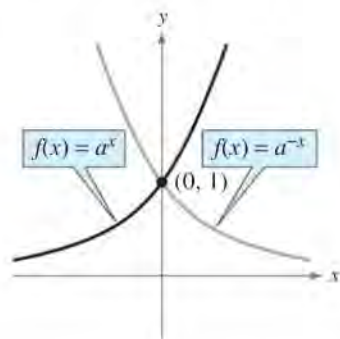
$$f(x) = \frac{1}{x}$$



Domain: $(-\infty, 0) \cup (0, \infty)$
 Range: $(-\infty, 0) \cup (0, \infty)$
 No intercepts
 Decreasing on $(-\infty, 0)$ and $(0, \infty)$
 Odd function
 Origin symmetry
 Vertical asymptote: y-axis
 Horizontal asymptote: x-axis

Exponential Function

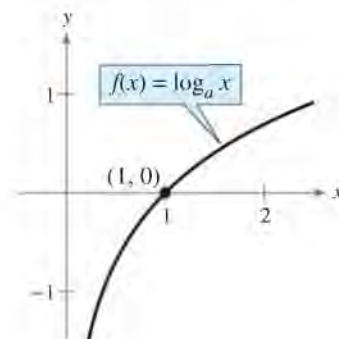
$$f(x) = a^x, a > 0, a \neq 1$$



Domain: $(-\infty, \infty)$
 Range: $(0, \infty)$
 Intercept: $(0, 1)$
 Increasing on $(-\infty, \infty)$
 for $f(x) = a^x$
 Decreasing on $(-\infty, \infty)$
 for $f(x) = a^{-x}$
 x-axis is a horizontal asymptote
 Continuous

Logarithmic Function

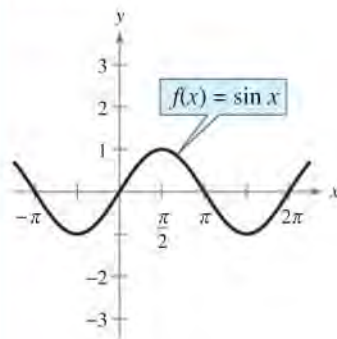
$$f(x) = \log_a x, a > 0, a \neq 1$$



Domain: $(0, \infty)$
 Range: $(-\infty, \infty)$
 Intercept: $(1, 0)$
 Increasing on $(0, \infty)$
 y-axis is a vertical asymptote
 Continuous
 Reflection of graph of $f(x) = a^x$
 in the line $y = x$

Sine Function

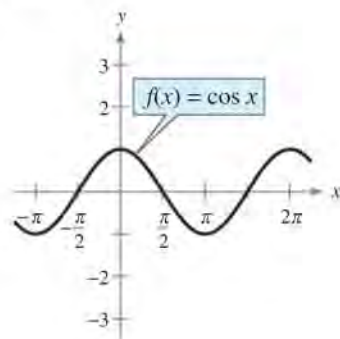
$$f(x) = \sin x$$



Domain: $(-\infty, \infty)$
 Range: $[-1, 1]$
 Period: 2π
 x-intercepts: $(n\pi, 0)$
 y-intercept: $(0, 0)$
 Odd function
 Origin symmetry

Cosine Function

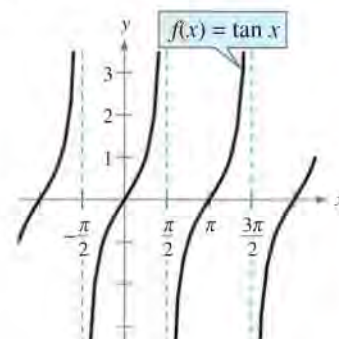
$$f(x) = \cos x$$



Domain: $(-\infty, \infty)$
 Range: $[-1, 1]$
 Period: 2π
 x-intercepts: $(\frac{\pi}{2} + n\pi, 0)$
 y-intercept: $(0, 1)$
 Even function
 y-axis symmetry

Tangent Function

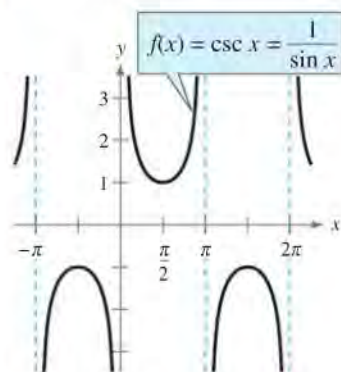
$$f(x) = \tan x$$



Domain: $x \neq \frac{\pi}{2} + n\pi$
 Range: $(-\infty, \infty)$
 Period: π
 x-intercepts: $(n\pi, 0)$
 y-intercept: $(0, 0)$
 Vertical asymptotes:
 $x = \frac{\pi}{2} + n\pi$
 Odd function
 Origin symmetry

Cosecant Function

$$f(x) = \csc x$$



Domain: $x \neq n\pi$

Range: $(-\infty, -1] \cup [1, \infty)$

Period: 2π

No intercepts

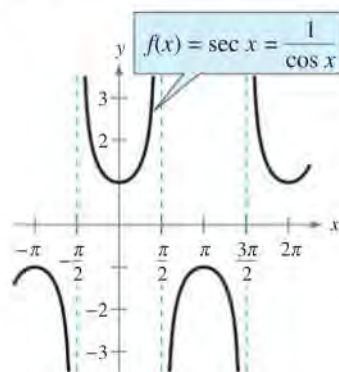
Vertical asymptotes: $x = n\pi$

Odd function

Origin symmetry

Secant Function

$$f(x) = \sec x$$



Domain: $x \neq \frac{\pi}{2} + n\pi$

Range: $(-\infty, -1] \cup [1, \infty)$

Period: 2π

y-intercept: $(0, 1)$

Vertical asymptotes:

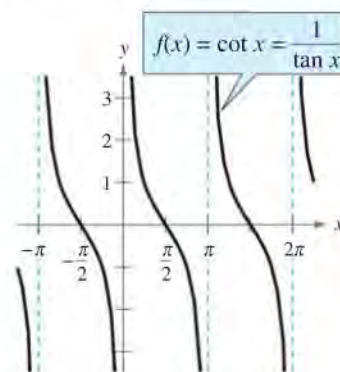
$$x = \frac{\pi}{2} + n\pi$$

Even function

y-axis symmetry

Cotangent Function

$$f(x) = \cot x$$



Domain: $x \neq n\pi$

Range: $(-\infty, \infty)$

Period: π

x-intercepts: $\left(\frac{\pi}{2} + n\pi, 0\right)$

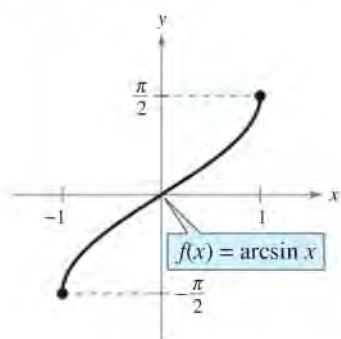
Vertical asymptotes: $x = n\pi$

Odd function

Origin symmetry

Inverse Sine Function

$$f(x) = \arcsin x$$



Domain: $[-1, 1]$

Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

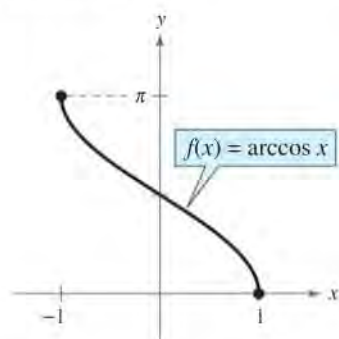
Intercept: $(0, 0)$

Odd function

Origin symmetry

Inverse Cosine Function

$$f(x) = \arccos x$$



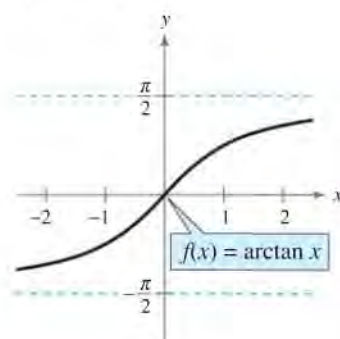
Domain: $[-1, 1]$

Range: $[0, \pi]$

y-intercept: $\left(0, \frac{\pi}{2}\right)$

Inverse Tangent Function

$$f(x) = \arctan x$$



Domain: $(-\infty, \infty)$

Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Intercept: $(0, 0)$

Horizontal asymptotes:

$$y = \pm \frac{\pi}{2}$$

Odd function

Origin symmetry